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prateeksrivastava-mac@prateeks-MacBook-Pro MFML_assignment_1 %
prateeksrivastava-mac@prateeks-MacBook-Pro MFML_assignment_1 %
prateeksrivastava-mac@prateeks-MacBook-Pro MFML_assignment_1 % python3 q2_matrix_decompositions.py
```

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=====
Q2: MATRIX DECOMPOSITIONS
=====
```

```
=====
Q2(a): LU Decomposition via Elementary Matrices
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```

```
Symmetric Positive Definite Matrix A:
```

```
[7.273, -1.072, 2.2, 1.14]
[-1.072, 13.258, 2.997, -5.569]
[2.2, 2.997, 16.266, 0.528]
[1.14, -5.569, 0.528, 9.252]
```

```
STEP-BY-STEP LU DECOMPOSITION USING ELEMENTARY MATRICES
=====
```

```
Starting matrix A:
```

```
:
[7.273, -1.072, 2.2, 1.14]
[-1.072, 13.258, 2.997, -5.569]
[2.2, 2.997, 16.266, 0.528]
[1.14, -5.569, 0.528, 9.252]
```

```
--- Step 1 ---
```

```
Operation: R2 = R2 - (-0.147) x R1
```

```
Elementary matrix E1:
```

```
:
[1.0, 0.0, 0.0, 0.0]
[0.147, 1.0, 0.0, 0.0]
[0.0, 0.0, 1.0, 0.0]
[0.0, 0.0, 0.0, 1.0]
```

```
Resulting matrix after E1 x A:
```

```
:
[7.273, -1.072, 2.2, 1.14]
[0.0, 13.1, 3.322, -5.401]
[2.2, 2.997, 16.266, 0.528]
[1.14, -5.569, 0.528, 9.252]
```

```
--- Step 2 ---
```

```
Operation: R3 = R3 - (0.302) x R1
```

```
Elementary matrix E2:
```

```
:
[1.0, 0.0, 0.0, 0.0]
[0.0, 1.0, 0.0, 0.0]
[-0.302, 0.0, 1.0, 0.0]
[0.0, 0.0, 0.0, 1.0]
```

```
Resulting matrix after E2 x A:
```

```
:
[7.273, -1.072, 2.2, 1.14]
[0.0, 13.1, 3.322, -5.401]
[0.0, 3.322, 15.601, 0.183]
[1.14, -5.569, 0.528, 9.252]
```

```
--- Step 3 ---
```

```
Operation: R4 = R4 - (0.157) x R1
```

```
Elementary matrix E3:
```

--- Step 3 ---

Operation: $R4 = R4 - (0.157) \times R1$

Elementary matrix E3:

```
:
[1.0, 0.0, 0.0, 0.0]
[0.0, 1.0, 0.0, 0.0]
[0.0, 0.0, 1.0, 0.0]
[-0.157, 0.0, 0.0, 1.0]
```

Resulting matrix after E3 x A:

```
:
[7.273, -1.072, 2.2, 1.14]
[0.0, 13.1, 3.322, -5.401]
[0.0, 3.322, 15.601, 0.183]
[0.0, -5.401, 0.183, 9.074]
```

--- Step 4 ---

Operation: $R3 = R3 - (0.254) \times R2$

Elementary matrix E4:

```
:
[1.0, 0.0, 0.0, 0.0]
[0.0, 1.0, 0.0, 0.0]
[0.0, -0.254, 1.0, 0.0]
[0.0, 0.0, 0.0, 1.0]
```

Resulting matrix after E4 x A:

```
:
[7.273, -1.072, 2.2, 1.14]
[0.0, 13.1, 3.322, -5.401]
[0.0, 0.0, 14.759, 1.552]
[0.0, -5.401, 0.183, 9.074]
```

--- Step 5 ---

Operation: $R4 = R4 - (-0.412) \times R2$

Elementary matrix E5:

```
:
[1.0, 0.0, 0.0, 0.0]
[0.0, 1.0, 0.0, 0.0]
[0.0, 0.0, 1.0, 0.0]
[0.0, 0.412, 0.0, 1.0]
```

Resulting matrix after E5 x A:

```
:
[7.273, -1.072, 2.2, 1.14]
[0.0, 13.1, 3.322, -5.401]
[0.0, 0.0, 14.759, 1.552]
[0.0, 0.0, 1.552, 6.847]
```

--- Step 6 ---

Operation: $R4 = R4 - (0.105) \times R3$

Elementary matrix E6:

```
:
[1.0, 0.0, 0.0, 0.0]
[0.0, 1.0, 0.0, 0.0]
[0.0, 0.0, 1.0, 0.0]
[0.0, 0.0, -0.105, 1.0]
```

Resulting matrix after E6 x A:

Resulting matrix after $E_6 \times A$:

```
:
[7.273, -1.072, 2.2, 1.14]
[0.0, 13.1, 3.322, -5.401]
[0.0, 0.0, 14.759, 1.552]
[0.0, 0.0, 0.0, 6.684]
```

=====

FINAL RESULT:

Lower triangular matrix L (with 1s on diagonal):

```
:
[1.0, 0.0, 0.0, 0.0]
[0.0, 1.0, 0.0, 0.0]
[0.0, 0.0, 1.0, 0.0]
[0.0, 0.0, 0.0, 1.0]
```

Upper triangular matrix U:

```
:
[7.273, -1.072, 2.2, 1.14]
[0.0, 13.1, 3.322, -5.401]
[0.0, 0.0, 14.759, 1.552]
[0.0, 0.0, 0.0, 6.684]
```

Note: L is constructed from inverses of elementary matrices

$L[i][j]$ = factor means we used E that subtracted factor* R_j from R_i

=====

SUMMARY - Final LU Decomposition Matrices:

Lower triangular matrix L:

```
[1.0, 0.0, 0.0, 0.0]
[-0.147, 1.0, 0.0, 0.0]
[0.302, 0.254, 1.0, 0.0]
[0.157, -0.412, 0.105, 1.0]
```

Upper triangular matrix U:

```
[7.273, -1.072, 2.2, 1.14]
[0.0, 13.1, 3.322, -5.401]
[0.0, 0.0, 14.759, 1.552]
[0.0, 0.0, 0.0, 6.684]
```

Number of elementary matrices used: 6

Verification $A = LU$:

Maximum difference: 0.0

Decomposition valid: YES

Logical Explanation (Q2a):

- LU decomposes $A = LU$ where L=lower triangular, U=upper triangular
- We use elementary matrices E_k to eliminate below-diagonal elements
- Each E_k performs: $\text{row}_i = \text{row}_i - \text{factor} \times \text{row}_j$
- After all eliminations: $E_n \dots E_2 \cdot E_1 \cdot A = U$ (upper triangular)
- Therefore: $A = E_1^{(-1)} \cdot E_2^{(-1)} \dots E_n^{(-1)} \cdot U = L \cdot U$
- L is built from inverses of elementary matrices (just flip signs)

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Q2(b): Cholesky Decomposition

=====

Using the same SPD matrix from Q2(a)

Cholesky factor L:

```
[2.6968, 0.0, 0.0, 0.0]
[-0.3976, 3.6194, 0.0, 0.0]
[0.8157, 0.9177, 3.8417, 0.0]
[0.4226, -1.4922, 0.4041, 2.5853]
```

Verification $A = L \cdot L^T$:
 Maximum difference: 0.0
 Decomposition valid: YES

Logical Explanation (Q2b):

- Cholesky works only for Symmetric Positive Definite (SPD) matrices
- It finds L such that $A = L \cdot L^T$ (like taking 'square root' of matrix)
- More efficient than LU: only need to compute lower triangle
- Formula: $L[i][j] = (A[i][j] - \sum L[i][k] \cdot L[j][k]) / L[j][j]$
- SPD guarantee ensures all diagonal elements remain positive

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Q2(c): QR Decomposition (Example with 5x3)

=====

Matrix A (5x3):

```
[-2.512, -3.369, 2.836]
[3.085, 1.256, 1.041]
[3.857, 2.591, -3.189]
[-3.498, -0.643, -1.147]
[0.757, -3.539, 1.866]
```

Orthonormal matrix Q:

```
[-0.3809, -0.4533, 0.1964]
[0.4677, -0.0639, 0.5526]
[0.5847, 0.1399, -0.6051]
[-0.5303, 0.2407, -0.3991]
[0.1148, -0.8443, -0.3614]
```

Upper triangular matrix R:

```
[6.5969, 3.3207, -1.6353]
[0.0, 4.6429, -3.6502]
[0.0, 0.0, 2.8458]
```

Verification $A = QR$:

Maximum difference: 0.0

$Q^T \cdot Q$ (should be identity):

```
[1.0, -0.0, -0.0]
[-0.0, 1.0, 0.0]
[-0.0, 0.0, 1.0]
```

Logical Explanation (Q2c):

- QR decomposes $A = Q \cdot R$ where Q =orthogonal, R =upper triangular
- Uses Gram-Schmidt: each column becomes orthogonal to previous ones
- Step 1: $u_j = a_j - \text{proj}(a_j \text{ onto } q_1) - \dots - \text{proj}(a_j \text{ onto } q_{j-1})$
- Step 2: $q_j = u_j / ||u_j||$ (normalize to unit length)
- $R[i][j]$ stores the coefficients: $R[i][j] = q_i \cdot a_j$
- Q has orthonormal columns: $Q^T \cdot Q = I$ (verified above)

=====

Q2(d): QR Decomposition with Random 7x5 Matrix

=====

Random 7x5 Matrix A:

```
[-2.83, -4.32, 4.7, 1.99, 7.61]
[4.14, -7.83, -7.6, -6.22, 0.61]
[-7.27, 0.83, 3.65, -3.23, 3.73]
[2.25, -4.47, -5.97, -2.76, -1.36]
[-5.79, 1.46, -2.07, 1.53, 4.59]
[5.54, 0.01, 6.57, -0.11, 5.63]
[5.14, -4.57, -4.91, -5.85, 4.81]
```

Q (7x5):

```
[-0.2144, -0.5512, 0.6257, 0.4416, 0.0909]
[0.3142, -0.6306, -0.1734, -0.129, -0.15]
[-0.5516, -0.206, 0.2077, -0.754, -0.1344]
[0.1705, -0.3646, -0.2393, 0.1048, -0.1356]
[-0.4394, -0.0827, -0.3074, 0.1563, 0.7437]
[0.4204, 0.2217, 0.6077, -0.1892, 0.334]
[0.39, -0.2591, -0.1195, -0.3854, 0.5179]
```

- Step 1: $u_j = a_j - \text{proj}(a_j \text{ onto } q_1) - \dots - \text{proj}(a_j \text{ onto } q_{j-1})$
- Step 2: $q_j = u_j / ||u_j||$ (normalize to unit length)
- $R[i][j]$ stores the coefficients: $R[i][j] = q_i \cdot a_j$
- Q has orthonormal columns: $Q^T \cdot Q = I$ (verified above)

```
=====
Q2(d): QR Decomposition with Random 7x5 Matrix
=====
```

Random 7x5 Matrix A:

```
[-2.83, -4.32, 4.7, 1.99, 7.61]
[4.14, -7.83, -7.6, -6.22, 0.61]
[-7.27, 0.83, 3.65, -3.23, 3.73]
[2.25, -4.47, -5.97, -2.76, -1.36]
[-5.79, 1.46, -2.07, 1.53, 4.59]
[5.54, 0.01, 6.57, -0.11, 5.63]
[5.14, -4.57, -4.91, -5.85, 4.81]
```

Q (7x5):

```
[-0.2144, -0.5512, 0.6257, 0.4416, 0.0909]
[0.3142, -0.6306, -0.1734, -0.129, -0.15]
[-0.5516, -0.206, 0.2077, -0.754, -0.1344]
[0.1705, -0.3646, -0.2393, 0.1048, -0.1356]
[-0.4394, -0.0827, -0.3074, 0.1563, 0.7437]
[0.4204, 0.2217, 0.6077, -0.1892, 0.334]
[0.39, -0.2591, -0.1195, -0.3854, 0.5179]
```

R (5x5 upper triangular):

```
[13.1846, -5.1712, -4.6701, -4.0734, -1.501]
[0.0, 9.8409, 6.5257, 5.8622, -5.2285]
[0.0, 0.0, 11.6619, 2.4764, 7.1893]
[0.0, 0.0, 0.0, 6.3388, -1.8777]
[0.0, 0.0, 0.0, 0.0, 8.0699]
```

Diagonal elements of R:

```
[13.1846, 9.8409, 11.6619, 6.3388, 8.0699]
```

Observation on Diagonal Elements of R:

ALL diagonal elements of R are STRICTLY POSITIVE.

This is a fundamental property of QR decomposition when columns are linearly independent.

Mathematical Explanation:

- Each $R[j][j]$ represents $||u_j||$ where u_j is the residual vector
- $u_j = a_j - \text{proj}_1 - \text{proj}_2 - \dots - \text{proj}_{j-1}$
- proj_k means the projection of column a_j onto orthonormal vector q_k
- Since columns are linearly independent, $u_j \neq 0$, so $||u_j|| > 0$

Physical Interpretation:

- Gram-Schmidt finds what's 'new' in each column after removing everything that can be expressed using previous columns
- If $R[j][j] = 0$, it would mean column j adds nothing new
- This would violate linear independence

Practical Consequences:

- R is invertible (no zero diagonal elements)
- QR decomposition succeeded without numerical breakdown
- Original matrix A has full column rank

Verification A = QR:

Maximum difference: 0.0

Decomposition valid: YES

Logical Explanation (Q2d):

- Applied same QR algorithm to larger 7x5 matrix
- Key insight: diagonal elements $R[j][j]$ reveal column relationships
- Positive diagonal $\rightarrow$ columns are linearly independent
- Each $R[j][j] = ||\text{residual}||$ after removing projections onto previous columns